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3 Intelligent Automation System Integration in Mobile and Industrial Robotics for Enhanced Performance and Efficiency

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Abstract

An automation system is a cohesive collection of machines and associated components to accomplish tasks or activities with minimal human involvement. These automation systems typically consist of actuators, controllers, sensors, and software that work together to automate a variety of tasks, including tool, manufacturing, healthcare, and control processes. Automation is designed to improve efficiency and reliability by reducing human error, streamlining workflows, and speeding up processes, which are used in a variety of industries including automotive, healthcare, aerospace, and logistics with improving accuracy, enabling complex tasks, and increasing overall productivity. Artificial intelligence (AI) and machine learning (ML) technologies stated together can enhance intelligent automation systems (IAS) to be more responsive and efficient because they can make decisions, learn from data, and adapt to changing circumstances and environments. The integration of AI in IAS has revolutionized industries by enabling these systems to perform tasks traditionally requiring human intelligence. In the field of robotics, AI has enabled the usage of IAS and robots for path planning, accident or collision avoidance, and trajectory tracking applications. However, the integration of IAS with AI brings several ethical concerns, including the risk of biased decision-making, which can result from algorithms reflecting existing prejudices or inequalities.

***Keywords*:** Industrial robotics, intelligent automation systems (IAS), mobile robotics, machine learning, tracking

3.1 Introduction

Industry development and automation processes are running together, and automation has become the foundation of industrial development in various operations like streamlining operations, decreasing human blunders, and enhancing efficiency. In the last few years, a new advancement is merged in the automation process, which can easily repeat the task and take the decision based on the previous task and have flexible operation [1]. It is termed "intelligent automation (IA)" and shows a wide scope and competence with current automation technology. Standard automation techniques include the machines to perform various tasks with the least involvement of humans, like assembly line production in which a machine is programmed to do a specific task precisely and repeatedly. This methodology has the drawback of managing with the diverse operating conditions and changing statistical values.

Intelligent innovation has benefited from artificial intelligence and machine learning technology, advanced and highly sensitive sensors to impart the automation system with more flexibility and decision-making capability. This automation system senses and evaluates the real-time data, predicts and adjusts the work conditions then makes decisions with the help of complex algorithms. The major components of IA are artificial intelligence, machine learning algorithms, and sensor-based automation systems. AI and machine learning algorithms allow the machine to learn from past experiences, enhance performance, and enable the machine to deal with a wide range of tasks [2]. In the manufacturing industry, IA technology has optimized production according to demand estimations, maintenance of machines and equipment based on their operation, and predications of failures and reliability. Besides this, IA is able to provide interface to communicate in the known natural language that human understands. It unlocks new opportunities in human-machine interaction, allowing operators to command the device through voice assistance or written commands. IA enhances the efficiency of operators and permits a high level of association between the device and the user.

Progressing from standard automation to AI-based automation has caused a large upgrade in the productivity and efficiency of the industry.

IA systems are tools that can perform multiple duties and can sense, analyze, and adjust according to the varying operating conditions. IA has the potential to transform the industry and can boost production in different sectors including manufacturing, healthcare, finance, and logistics.

Robotics takes sensible automation a step further by means of adding physical abilities to the equation [3]. Robots are essentially physical machines that may be programmed to perform tasks autonomously. They may be geared up with sensors, cameras, and different technologies that allow them to engage with their environment and make decisions based totally on the information they acquire [4]. Robotics has programs in loads of industries, from manufacturing and logistics to healthcare and agriculture. One of the important benefits of robotics is its capability to carry out tasks that are too risky or hard for human beings [5–7]. For instance, robots may be used to address dangerous substances, paintings in extreme environments, or perform repetitive responsibilities with precision and consistency. In manufacturing, robots are used to gather products, handle substances, and perform high-quality manipulate inspections. Robots are evaluating laboratory data and performing patient care and surgeries in the medical sector [8–10].

This research is aimed at representing the performance of mobile and industrial robots. In Section 1, the construction of an anthropomorphic robot and its motion range is examined. It can be successfully used in different industries for handling heavy items, assembly, and production line automation. This research explores the control technology and articulations of the mechanical framework of anthropomorphic robots. The robot is directed to a particular path for simulating the operation for arranging and selecting goods in the manufacturing process. In Section 3.2, the construction of a mobile autonomous robot is highlighted for a specific area. The significance of the designed robot in various scenarios like logistics, operation in harsh environments, and health care is discussed. A navigation algorithm is uploaded in the robot to enable it for sensing the surroundings and coordinate the orientation of motion. Further, the control system unit is designed to ensure the motion of the robot along a predefined path effectively and safely.

With the help of designed robots, technical and conceptual concerns are undertaken. Thus, the primary goal is to investigate the possibilities of mobile and anthropomorphic robots and apply them to certain settings like industrial and mobile robotics. It is believed that this research will offer a thorough summary of the actions taken, the outcomes attained, and the potential directions for further research and development of sophisticated robots.

3.2 Industrial Robotics

Industrial robotics has been a game-changer in the production industry, automating processes and pushing the boundaries of efficiency, productivity, and scalability. These state-of-the-art industrial machines, often equipped with superior sensors, controllers, and software, perform a wide range of responsibilities with precision and consistency, in the long run transforming the manner the way products are made. It offers various advantages like automation of repeated hazardous operations, mitigating human error, improving the safety of operators, no time barrier to work, improved production, and reduced cost [11].

In addition to this, industrial robots are flexible and more capable of performing various tasks including packaging, assembly, painting, and welding. It is easy to reprogram a robot and reconfigure it to align with new production requirements. It is a cost-effective way to make flexible and agile tools for the advanced production industry. The industrial robot also benefits from the improvement of the quality of production. The robot follows the standards and requirements with high consistency and precision. It is difficult to gain this level of accuracy, and standard of expectations for the manual operators [12]. Industrial robot is featured with the advanced and complex ML- and AI-based computational techniques that enable it to get knowledge from past experiences through the years and increase production capacity.

3.3 Anthropomorphic Robot: Bridging the Gap Between Humans and Machines

Anthropomorphic or humanoid robots are developed to resemble and act similarly to human postures, movements, and behaviors. These are integrated with the human-like body parts such as legs, arms, and heads that allow it to behave similarly to the human body. The primary focus of designing this kind of robot is to reduce the gap between machines and humans and ensure a natural and comfortable interaction platform [13, 14]. Humanoid robots are essentially required for human-and-machine collaboration in the entertainment industry, healthcare sector, and rehabilitation centers. It is a challenging task to establish natural and living body movements in a machine. Advanced engineering and technologies are required to design anthropomorphic robot parts like sensors, actuators, and artificial muscles and enable a robot to move in a realistic manner [15].

In the medical sector, anthropomorphic robots are used to assist doctors in patient care, rehabilitation, and surgeries by providing personal and interactive support. In the entertainment industry, humanoid robots attract the audience in theme parks. In academics, researchers are exploring opportunities to enhance the participation of robots in the dayto-day life of human beings. With the advancement of technology, robots are playing an important role in the development of society. Robots are interfering with the way of living and work of humans [16].

Among the various components of a robot, the controller is the key component to operate it effectively. For example, to allow a robotic arm to perform different tasks with accuracy, safety, and skilled way. The major functions that are performed by a controller in a robotic arm are movement control, feedback control, trajectory planning, collision avoidance, coordination of multiple joints, and integration with human inputs. For designing controllers for the robotic arm, the first modeling is completed, which needs an understanding of the device's mathematical and physical properties. The basis lies in two key models: kinematic and dynamic. Kinematics analyzes motion (position, velocity, acceleration) without considering forces, and dynamics explores the relationship among forces and resulting motion. Both are vital for robotic manipulators, as they describe

the complex interplay of movement and forces that govern their operation. Often used for instructional features, a 3-degree-of-freedom (3-DOF) manipulator serves as a simplified example to illustrate these concepts.

3.3.1 Forward Kinematics

Forward kinematics deals with determining the end-effector's position and orientation based on the joint angles. For a 3-DOF manipulator, the Denavit-Hartenberg (DH) parameters outline the geometry of every joint and link [17]. Let us designate the joint angles as $\theta_1,\theta_2,\theta_3$, and the corresponding link lengths as l_1,l_2,l_3. The DH parameters for each joint are denoted as $(a_i,\alpha_i,d_i,\theta_i)$. The transformation matrix from the base frame to the end-effector frame can be obtained by multiplying the individual transformation matrices for each joint. The transformation matrix for each joint can be calculated as follows:

At Joint 1:

$$T_1 = \begin{bmatrix} c\,\theta_1 & -s\,\theta_1 & 0 & 0 \\ s\,\theta_1 & c\,\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.1}$$

At Joint 2:

$$T_2 = \begin{bmatrix} c\,\theta_2 & -s\,\theta_2 & 0 & l_2\,c\,\theta_2 \\ s\,\theta_2 & c\,\theta_2 & 0 & l_2\,s\,\theta_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.2}$$

At Joint 3:

$$T_3 = \begin{bmatrix} c\,\theta_3 & -s\,\theta_3 & 0 & l_3\,c\,\theta_3 \\ s\,\theta_3 & c\,\theta_3 & 0 & l_3\,s\,\theta_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.3}$$

where *c* and *s* represents the *Cos* and *Sin* functions, respectively. The overall transformation can be calculated by taking the product of base frame matrix and end-effector frame matrix.

$$T_{\text{end}} = T_1 T_2 T_3 \tag{3.4}$$

This matrix represents the forward kinematics of the 3DOF robotic arm, mapping joint angles $\theta_1,\theta_2,\theta_3$ to the end-effector position and orientation.

3.3.2 Inverse Kinematics

For a 3-DOF robotic arm, the inverse kinematics problem involves finding the joint angles $\theta_1,\theta_2,\theta_3$ given the desired position and orientation of the end-effector. Let us denote the desired position of the end-effector as (x_d,y_d,z_d) and the desired orientation as a rotation matrix R_d. The inverse kinematics equations can be formulated as follows:

The position of the end-effector can be determined using the forward kinematics equations:

$$\begin{bmatrix} x_d \\ y_d \\ z_d \end{bmatrix} = \begin{bmatrix} f_{11}(\theta_{123}) & f_{12}(\theta_{123}) & f_{13}(\theta_{123}) \\ f_{21}(\theta_{123}) & f_{22}(\theta_{123}) & f_{23}(\theta_{123}) \\ f_{31}(\theta_{123}) & f_{32}(\theta_{123}) & f_{33}(\theta_{123}) \end{bmatrix} \begin{bmatrix} l_1 \\ l_2 \\ l_3 \end{bmatrix} \tag{3.5}$$

where f_{ij} are functions representing the forward kinematics equations and θ_{123} is $\theta_1,\theta_2,\theta_3$.

The orientation of the end-effector can be determined by comparing the desired rotation matrix R_d with the rotation matrix obtained from the joint angles $\theta_1,\theta_2,\theta_3$.

3.3.3 Forward Dynamics

Forward dynamics is the motion of a robot arm with the forces and torques applied to it. With a 3DOF manipulator, the dynamic equations can be complex and are usually derived using the Lagrange method. The equations for a 3DOF manipulator can be written as follows:

$$\tau = M(q)\ddot{q} + C(q,\dot{q})\dot{q} + G(q) \tag{3.6}$$

where

τ is the vector of joint torques,

$M(q)$ is the mass matrix,

$C(q,\dot{q})$ is the vector of Coriolis and centrifugal forces,

$G(q)$ is the vector of gravitational forces,

q is the vector of joint positions,

$\dot{q}$ is the vector of joint velocities,

$\ddot{q}$ is the vector of joint accelerations.

Mass matrix $M(q)$ represents the inertia properties of the robot and depends on the configuration of the robot (joint angles). Coriolis and centrifugal forces $C(q,\dot{q})$ arise due to the motion of the robot and are a function of both joint positions and velocities. Gravitational forces $G(q)$ arise due to the effects of gravity on the robot's links and depend on the joint positions.

3.3.4 Inverse Dynamics

Inverse dynamics calculates the torques required at the joints to achieve a desired motion. The inverse dynamics equations can be written as:

$$\tau = M(q)\ddot{q} + C(q,\dot{q})\dot{q} + G(q) + \tau_{ext} \tag{3.7}$$

where τ_{ext} is the vector of external torques applied to the manipulator. Both forward and inverse dynamics are key concepts in robot control

and are used in a variety of applications, including robot simulation, motion planning, and controller design.

3.3.5 Denavit-Hartenberg Method

In robotics to simulate the kinematics of robotic manipulators, the Denavit-Hartenberg (DH) method is used. The DH method consists of four parameters for each joint, i.e., link length (a), link twist (α), link offset (d), and joint angle (θ). By properly assigning these parameters to each joint and using them to generate transformation matrices, the DH method can determine the position and orientation of the end-effector with respect to the robot-based frame. The importance of this information for robot control and path planning makes the DH method a key tool in robot design.

Consider an anthromorphic robot as shown below: -

The four DH method-specific principles were followed for setting the frames of each joint in the robot. The four DH parameters were then determined using Figure 3.1, which depicts frame $\boldsymbol{n}$'s position in relation to frame $\boldsymbol{n}$ − 1, and entered into a parameter table as indicated in Table 3.1.

Rotary joint angle (θ_i), sliding joint variable (d_i), joint offset distance (a_i), and angle between the two adjacent z-axes (α_i) make up the four parameters. The next step involved computing each homogeneous transformation matrix needed to explain the change from frame 3 to frame 0 by changing each entry in each row of Table 3.1 into the corresponding matrices.

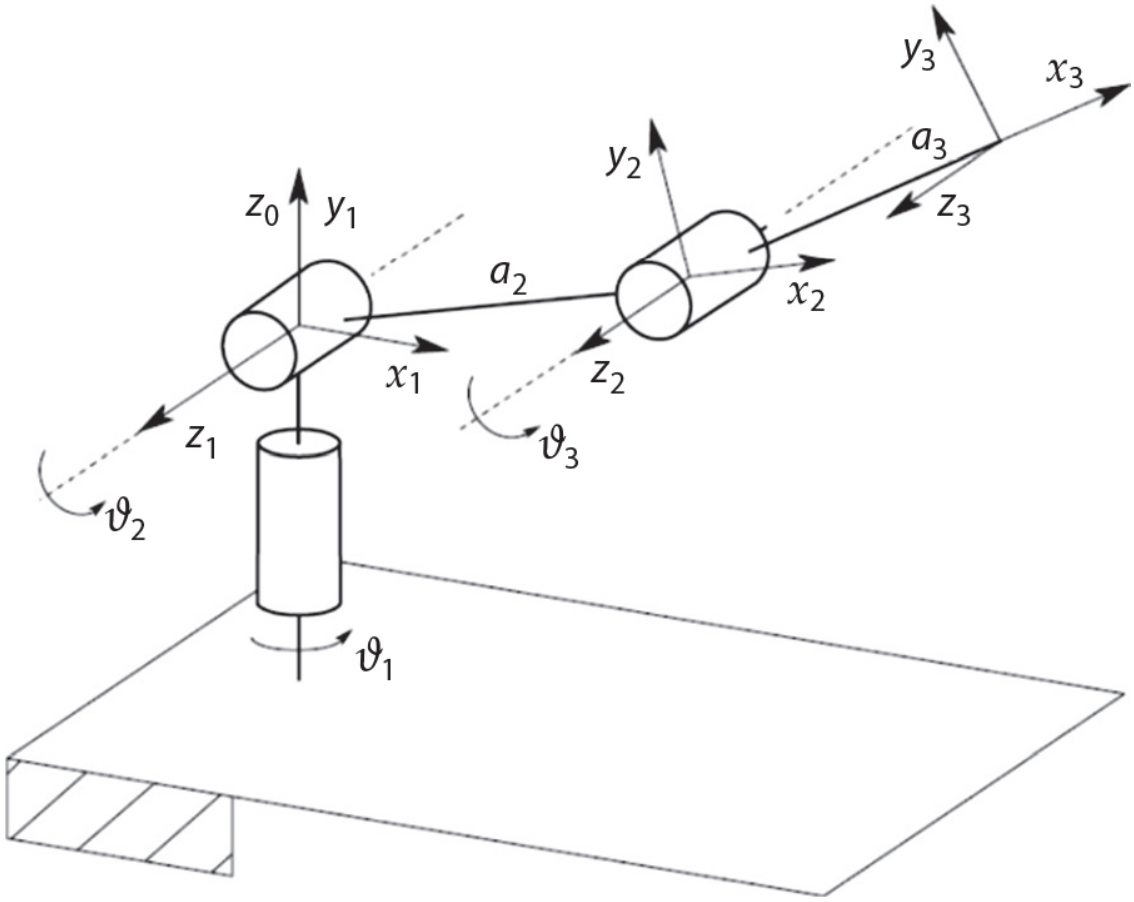


Figure 3.1 Kinematic diagram of 3-DOF anthropomorphic manipulator.

Table 3.1 DH parameters for an anthropomorphic robot arm.

Link	a_i	α_i	d_i	θ_i
1	0	$\frac{\pi}{2}$	0	θ_1
2	a_2	0	0	θ_2
3	a_3	0	0	θ_3

$$T_1^0 = \begin{bmatrix} c(\theta_1) & -s(\theta_1)\,c(\alpha_1) & s(\theta_1)\,s(\alpha_1) & a_1\,c(\theta_1) \\ s(\theta_1) & c(\theta_1)\,c(\alpha_1) & -c(\theta_1)\,s(\alpha_1) & a_1\,s(\theta_1) \\ 0 & s(\alpha_1) & c(\alpha_1) & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (3.8)$$

The T_0^1 matrix represents the transformation from frame 1 to frame 0 in terms of DH parameters θ_1, d_1, a_1 , and α_1. Similarly, T_2^1 and T_3^2 are calculated.

To write the complete transformation matrix from frame 3 to frame 0 using the DH parameters, multiply the individual transformation matrices from each frame to the next. The general formula for the transformation matrix from frame 3 to frame 0 becomes:

$$T_3^0 = T_1^0 \times T_2^1 \times T_3^2 \quad (3.9)$$

where T_i^j represents the transformation matrix from frame *j* to frame *i*.

3.4 Case Study

3.4.1 Kinematics of a 3-DOF Anthropomorphic Manipulator

Consider an anthropomorphic robot whose structure is modeled by DH parameters given in Table 3.2 and in which the reference systems (0) and (b) are linked by the roto-translational matrix given below.

$$\widehat{R}_b^0 = \begin{bmatrix} 1 & 0 & 0 & 0.5 \\ 0 & 1 & 0 & 0.5 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.10}$$

Table 3.2 DH parameters for an anthropomorphic robot arm.

Link	a_i	α_i	d_i	θ_i
1	0	$\frac{\pi}{2}$	0	θ_1
2	0.9	0	0	θ_2
3	0.9	0	0	θ_3

Consider the following points

$$P_1 = \begin{bmatrix} 0.8 \\ 0.8 \\ 0.5 \end{bmatrix}; P_2 = \begin{bmatrix} 1.2 \\ 0.8 \\ 0.5 \end{bmatrix}; P_3 = \begin{bmatrix} 1.0 \\ 1.2 \\ 0.5 \end{bmatrix} \tag{3.11}$$

Now, we need to determine the time trends of the joint variables (position and speed) so that the origin of the Coordinate Reference System (CRS is a specific point in a three-dimensional space used as a reference for measuring positions of objects or points.): -

1. Describes a triangle according to the sequence P1 → P2 → P3 → P1
2. Describes a circle passing through the points P1 → P2 → P3 → P1

In both cases, the total travel time of the bend must be 40 seconds.

To determine the time trends of the joint variables (position and speed) so that the origin of the CRS follows a specific trajectory, we would need to calculate the joint angles and velocities required to move the end-effector

(or any specified point) of a robotic manipulator along that trajectory. This involves kinematic analysis and possibly dynamic analysis of the robot, depending on the complexity of the trajectory and the robot's dynamics. A comprehensive analysis of the robotic arm's motion along the triangular and circular paths, including joint angles, end-effector positions, and velocities is shown in Figure 3.2.

For triangular path, as shown in Figure 3.3, it can be seen as the union of three straight paths, three segments to be precise. A segment can be parameterized as follows:

$$P(t) = P_1 + \lambda(t)[P_2 - P_1] \tag{3.12}$$

To get lambda λ, we use the polynomial function, to which we pass the sigma parameter; σ is a dimensionless independent variable that varies between 0 and 1. Points P1, P2, and P3, i.e., the vertices of the triangle, vector L, containing the lengths of the arms (extrapolated from the DH table) and the execution times of the movement (T1, T2, T3, and T4), are also calculated. The execution times are obtained by dividing the total time that the robot must take to travel the trajectory into three parts, approximately 13.33 sec. The lambda_d variable is also calculated, which is the derivative of lambda, used for calculating the joint speeds. We then proceed by traversing the section from P1 → P2 at each iteration the position of the end-effector is calculated using function $P(t)$, the value of the joint variables using inverse kinematics and the value of the joint speed. For the calculation of the speeds, the inverse differential kinematics is applied. In particular, the following relationship is exploited:

$$\dot{P} = J(Q) \cdot \dot{Q} \tag{3.13}$$

$$\dot{Q} = J(Q)^{-1} \cdot \dot{P} \tag{3.14}$$

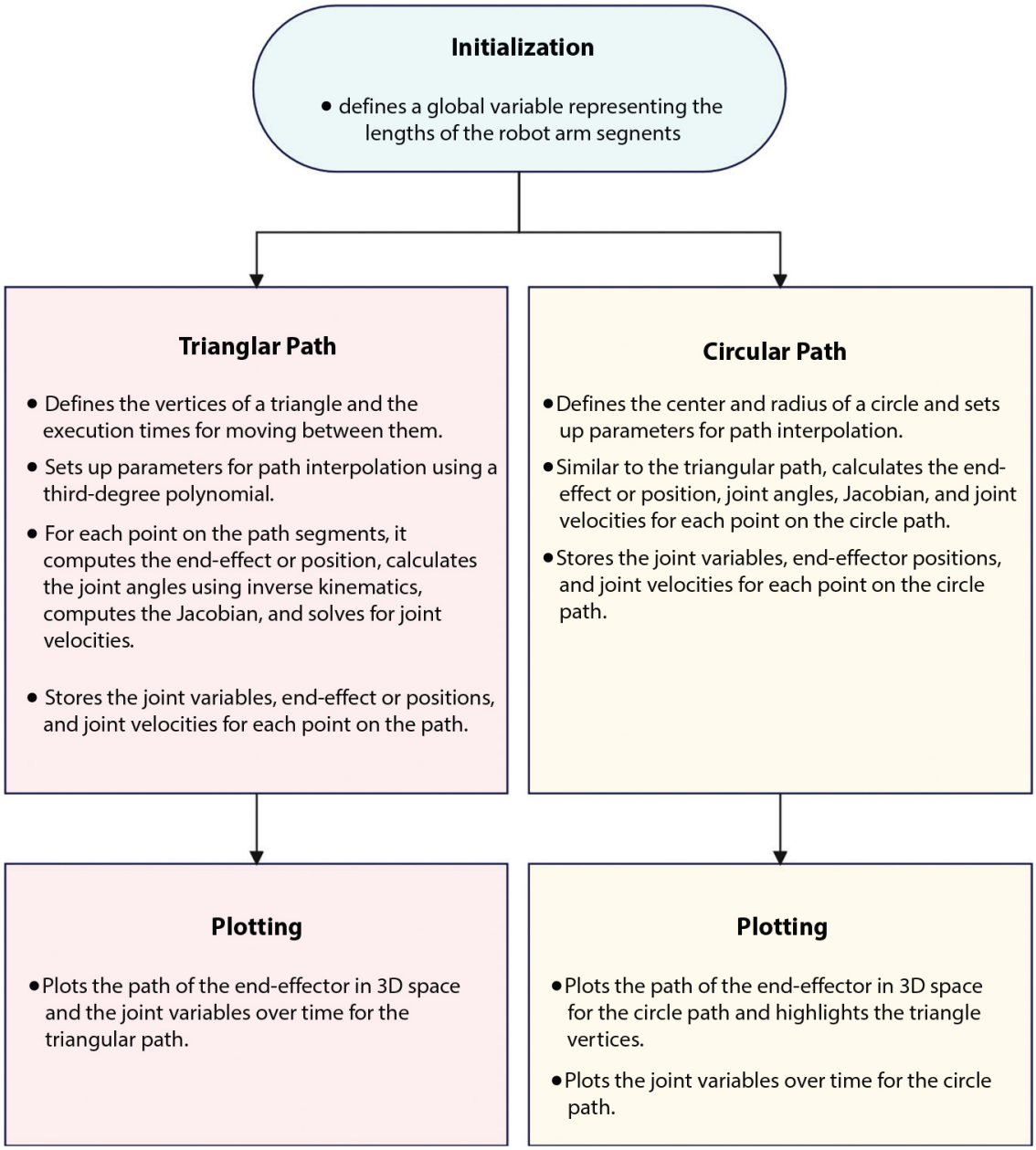


Figure 3.2 A comprehensive analysis of the robotic arm's motion along the triangular and circular paths, including joint angles, end-effector positions, and velocities.

The paths P2 → P3 and P3 → P1 are calculated in the same way.

Finally, the temporal trends of the joint variables can be visualized, as shown in Figure 3.4 and Figure 3.5, after having parameterized the time and converted the angles expressed in radians into angles expressed in degrees:

For a circular path, as shown in Figure 3.6, it is necessary to parameterize the path to be followed. The generic equation of the circle in the plane is

given by:

$$P = P_c + R \cdot \begin{bmatrix} cos(\theta) \\ \sin(\theta) \end{bmatrix}, \quad 0 \le \theta \le 2\pi \tag{3.15}$$

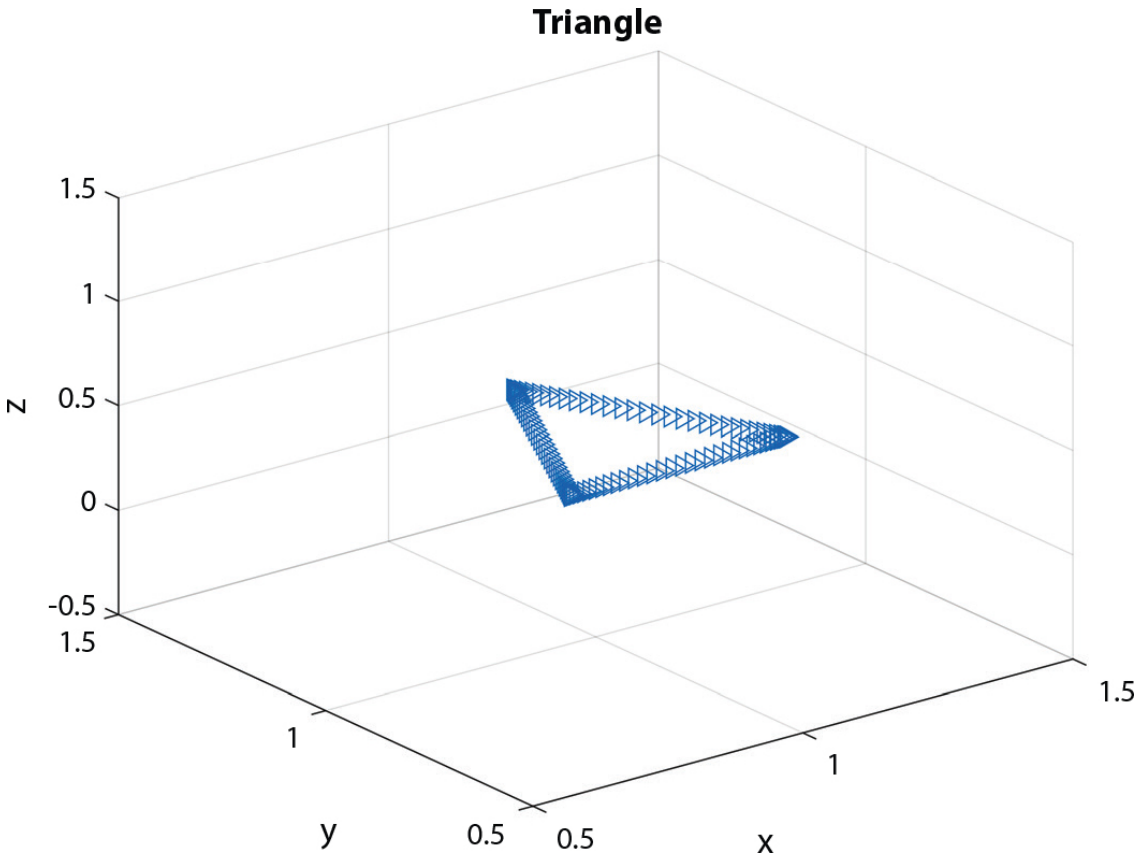


Figure 3.3 Triangular path.

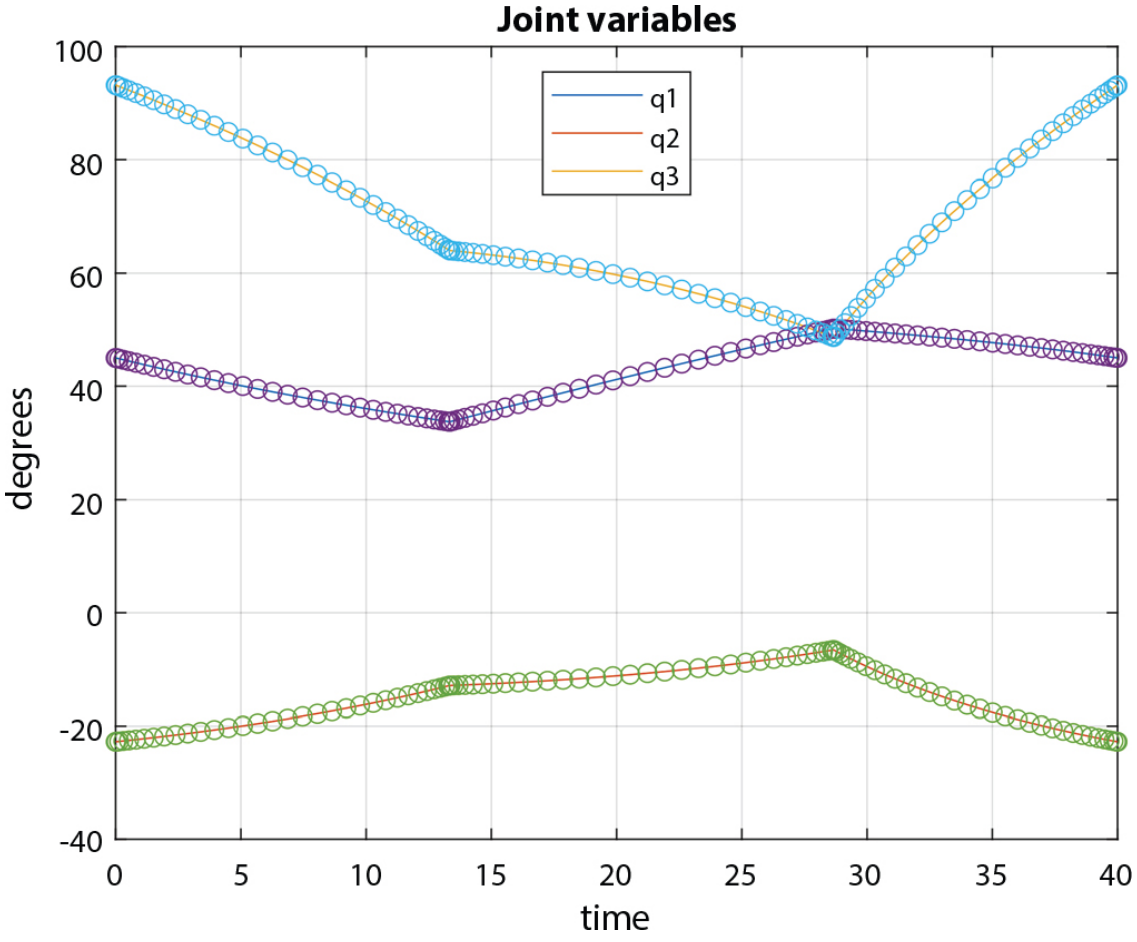


Figure 3.4 Joint variables for triangular path.

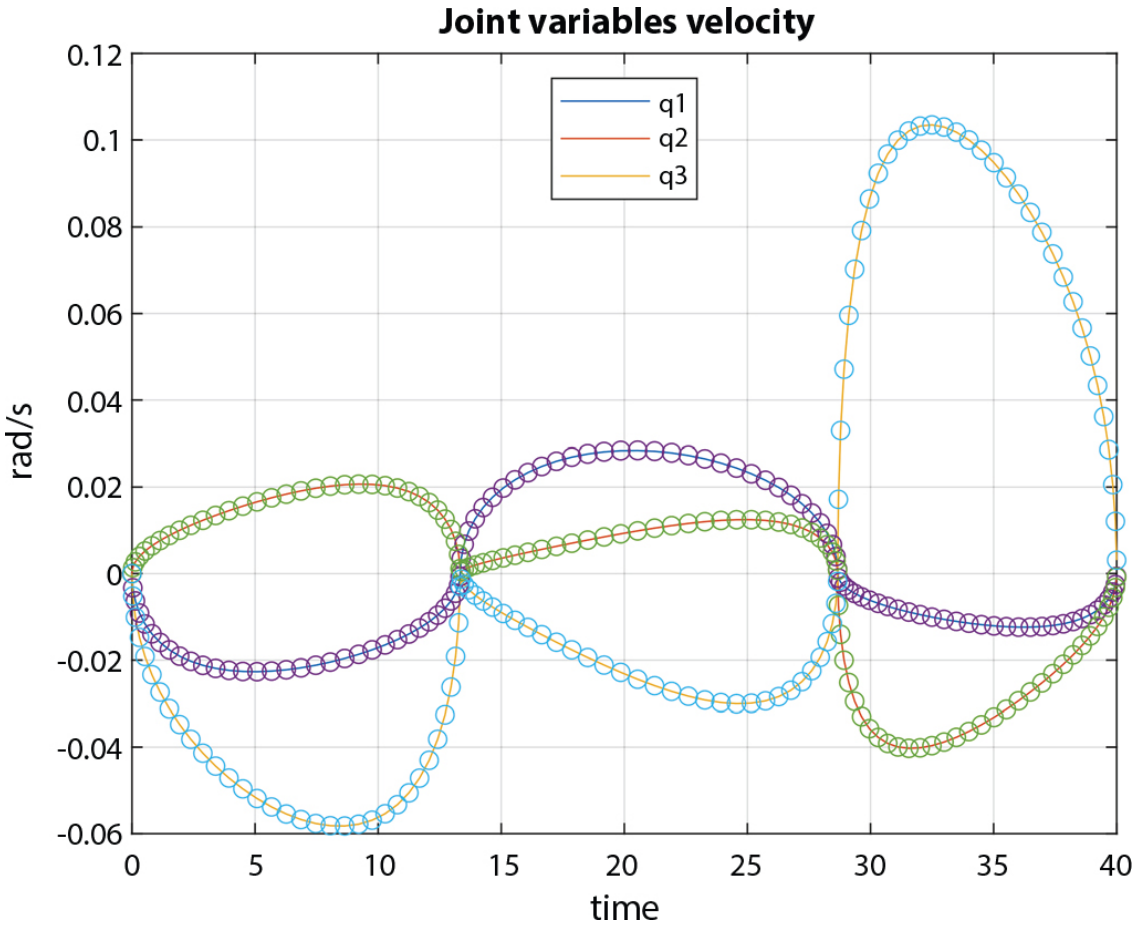


Figure 3.5 Joint variables velocity for triangular path.

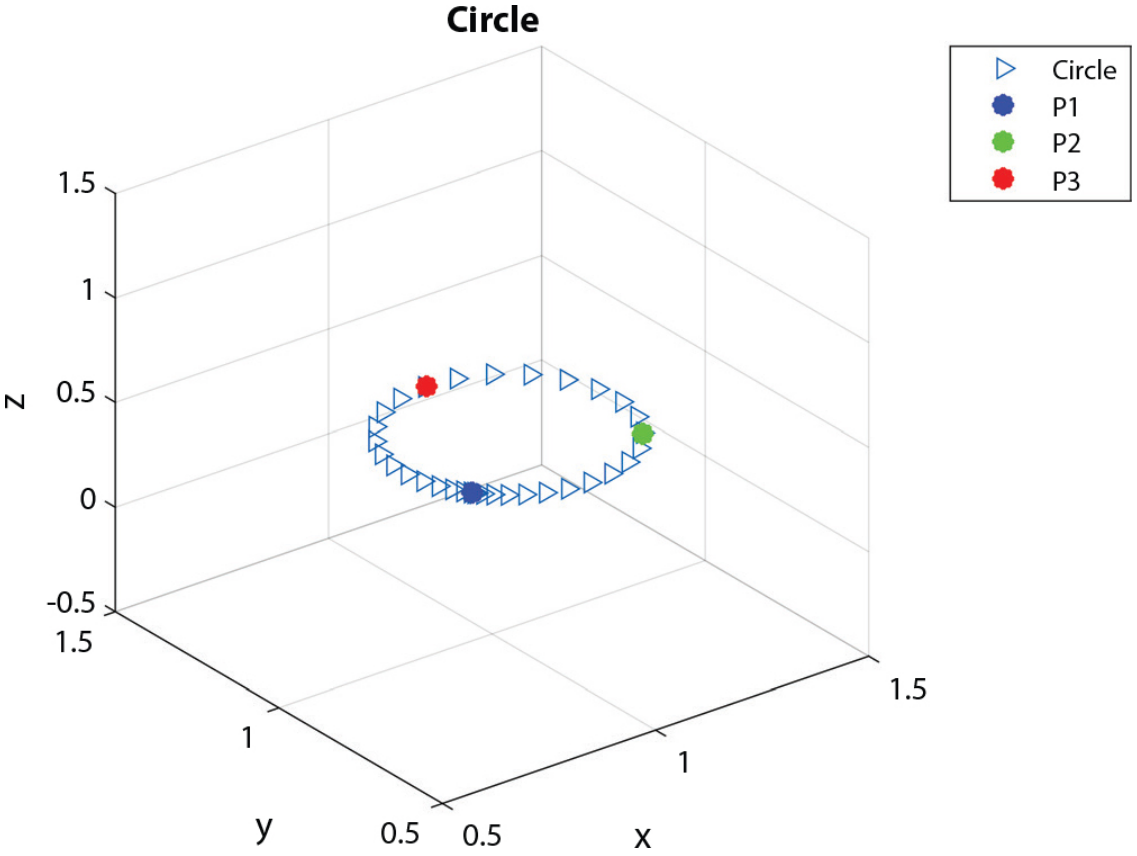


Figure 3.6 Circular path.

In the case under examination, extending in $\mathbb{R}^3$:

$$\begin{cases} P_x = P_{cx} + R \cdot \cos(\theta) \\ P_y = P_{cy} + R \cdot \sin(\theta) \\ P_z = P_{cz} \end{cases} \tag{3.16}$$

$$P = P_c + R \cdot \begin{bmatrix} cos(\theta) \\ \sin(\theta) \\ 0 \end{bmatrix}, \quad with\, 0 \le \theta \le 2\pi \tag{3.17}$$

From the equation of the circumference that passes through P1, P2 and P3, we obtain the center and the radius: P_c = (1.0,0.95,0.5), r = 0.25

Before proceeding with the implementation, it is important to specify that the path to be followed must start from point P1; therefore, the range in which theta varies must be: $\theta_1 \le \theta \le \theta_1 + 2\pi$. Using the principles of trigonometry, we get that $\theta_1 \cong 3.5rad$

Furthermore, it is necessary to express theta as a function of λ: $\theta = \theta_1 + \lambda [\theta_2 - \theta_1]$

In the present case: $\theta = \theta_1 + \lambda[\theta_1 + 2\pi - \theta_1] = 3.8 + \lambda \cdot 2\pi$

Analogously to the previous case, the section P1 → P2 → P3 → P1 is covered, and the joint variables, as shown in Figure 3.7 and Figure 3.8, are obtained by applying the inverse kinematics and the inverse differential kinematics:

3.4.2 Mobile Robotics

Designing, creating, and deploying robots that can move autonomously or semi-autonomously in varying surroundings is known as mobile robotics. These robots can sense their environment, make judgments, and navigate through challenging situations because they are equipped with sensors, actuators, and sophisticated algorithms. Because they can perform jobs that might be hazardous, tedious, or challenging for human beings, mobile robots have the capacity to completely transform some industries.

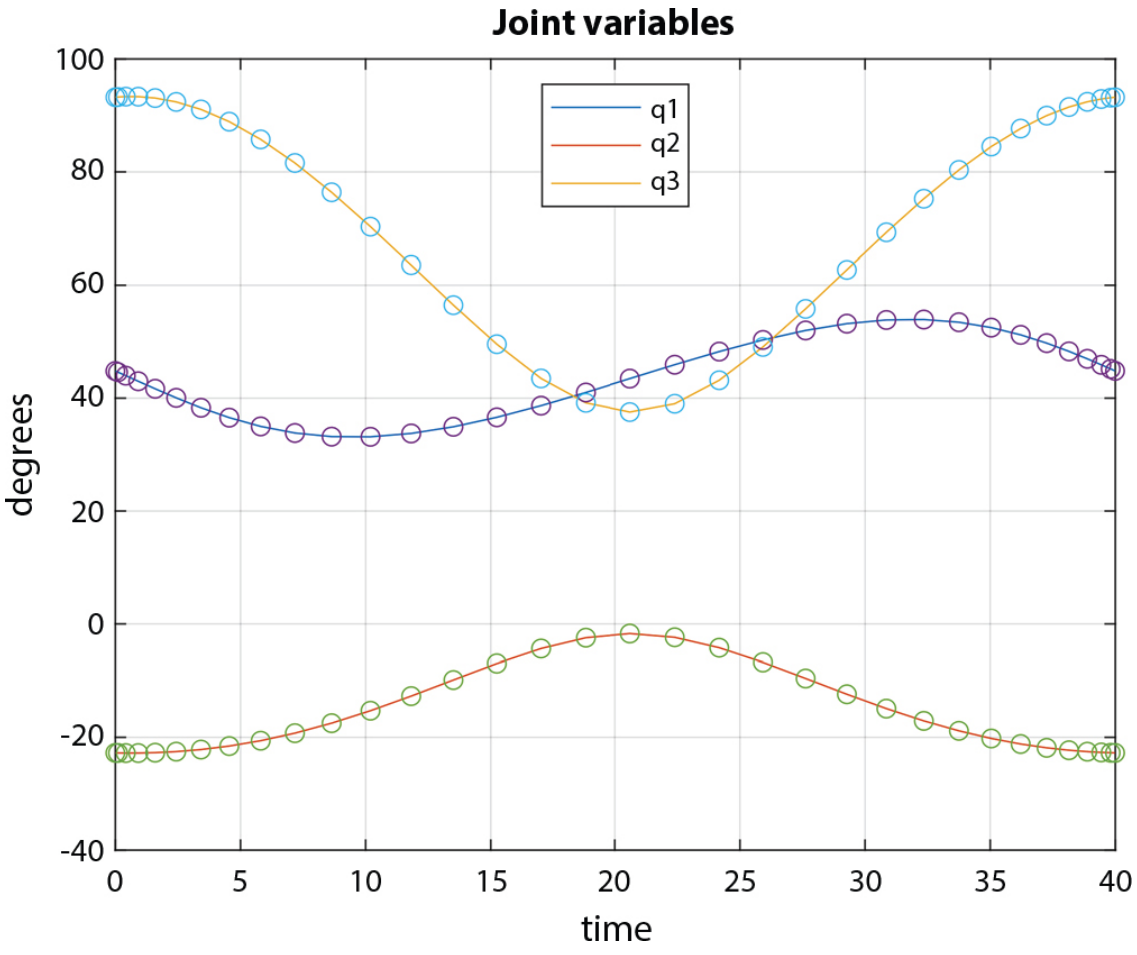


Figure 3.7 Joint variables for circular path.

Figure 3.8 Joint variables velocity for circular path.

1. **Locomotion Mechanisms:** Mobile robots can use a variety of locomotion mechanisms like wheels, tracks, legs, and aerial structures. Each mechanism has its advantages and limitations, depending on the terrain and application. Recent advances in locomotion include the development of biologically inspired soft robots that are flexible and adaptable to unexpected environments.

2. **Sensing and Perception:** Sensors provide information about the robot's surroundings such as cameras [18], LiDAR [19], and inertial measurement units (IMUs). They improve robots' perception capabilities by incorporating ML techniques, such as deep learning for object recognition and scene understanding in mobile robots.
3. **Localization and Mapping:** Simultaneous Localization and Mapping (SLAM) algorithms enable robots to navigate and function autonomously inside maps of their environments at the same time as approximating their very own positions in those maps.
4. **Path Planning and Control:** Path planning algorithms determine the ideal path for a robot to move from the present position to the destination or target while avoiding obstacles [20]. These algorithms consider various factors such as robot kinematics, environment, size and number of obstacles, and task requirements. Control algorithms ensure that the robot follows the planned path more accurately and efficiently, taking into account its strengths and limitations [21].
5. **Human-Robot Interaction:** It focuses on how humans and robots can collaborate and communicate effectively. Advances in these communications include sensors, gesture recognition, and the use of open language to facilitate interaction between humans and robots.

3.4.3 Obstacle Avoidance

In this case study, we will create a code that can navigate a robot from a start point to a goal point with a specific final orientation after creating an environment with obstacles. To accomplish this, the following control mechanism, as shown in Figure 3.9, needs to be put into place:

- The task can be selected by the user, who can also set the robot's orientation, starting position, and end position.
- The planners that the user can choose from include Voronoi Diagrams, Discrete Potential Fields, Artificial Potential Fields, and complete Visibility Graphs.
- The robot can be guided along the course by selecting one of three controller types: linear, nonlinear, or input/output linearization.
- The system is intended for mobile robots with differential drives or unicycles.

- The surroundings are preset.
- It is not necessary to implement the sensor block.

To do this, the following control scheme must be implemented:

Mobile Robot Block

This block defines the robot model (unicycle model) and sets the robot's diameter as a key structural parameter. In this case study, we have chosen the unicycle model because it simplifies the robot's motion into a single-wheel model which makes it easier to analyze and control the robot. In the unicycle model, the control signals are determined separately in the Controller block which allows for a more focused analysis and design of control strategies specific to the unicycle model. This designing ensures precise and effective control of the robot's motion. The unicycle model provides a good approximation for many wheeled robots, especially in environments where the terrain is relatively flat and smooth.

The unicycle model describes the robot's motion using two kinematic equations: one for linear velocity v and one for angular velocity ω. These equations relate the robot's velocities to its current orientation θ and the control inputs, which are typically linear velocity v and angular velocity ω commanded by the controller.

Figure 3.9 Mobile robot control scheme.

The state of the unicycle model can be represented by the robot's position x, y and orientation θ. The state update equations for the unicycle model can be written as follows:

$$(3.18)$$

$$(3.19)$$

$$(3.20)$$

where , , and are the rates of change of x, y , and θ, respectively.

These equations can be integrated over time to determine the robot's position and orientation at any given time, given an initial position and ori-

entation. It is also assumed that the robot moves through the combination of two actions: 1. Linear velocity, indicating the forward speed of the robot along its orientation. 2. Rotational velocity, indicating the speed at which the robot rotates around its center. It is essential to emphasize that the model in question is a simplification of reality and does not account for factors such as wheel slippage, internal dynamics of the robot, or the presence of obstacles in the environment.

Environment Block

This block defines the predefined environment in which the robot will operate, including defining obstacle locations within the environment. A detailed description of the created environment is discussed below:

Room Dimensions: The room is rectangular with a width of 100 units and a height of 100 units. It is defined by four corners: (0, 0) (bottom-left), (100, 0) (bottom-right), (0, 100) (top-left), and (100, 100) (top-right).
Obstacles: Several obstacles are placed within the room, each defined by four points representing its corners in the following format: [southwest; south-east; north-west; north-east].

1. Obstacle 1: Located at (10, 35) (south-west), (30, 35) (south-east), (10, 85) (north-west), and (30, 85) (north-east).
2. Obstacle 2: Located at (20, 10) (south-west), (60, 10) (south-east), (20, 25) (north-west), and (60, 25) (north-east).
3. Obstacle 3: Located at (50, 40) (south-west), (65, 40) (south-east), (50, 55) (north-west), and (65, 55) (north-east).
4. Obstacle 4: Located at (70, 15) (south-west), (90, 15) (south-east), (70, 45) (north-west), and (90, 45) (north-east).
5. Obstacle 5: Located at (65, 70) (south-west), (85, 70) (south-east), (65, 85) (north-west), and (85, 85) (north-east).

Initial Position, Goal, and Orientation Selection

This block allows the user to input the starting point, goal, and final orientation for the robot, as shown in Figure 3.10. Orientation θ can be selected accordingly. In our case study, we have selected starting point as [1,1], goal as [100,100], and orientation as θ = 45 degree.

Figure 3.10 Room with obstacles, starting point, and goal (final point).

Planner Block

This block provides options for selecting a path planning technique (e.g., artificial potential fields, discrete potential fields, Voronoi diagrams, and visibility graphs) to generate a trajectory for the robot.

1. **Artificial Potential Fields:** Treats the robot as a point in a field where attractive forces pull it toward the goal and repulsive forces push it away from obstacles. An example of this is shown in Figure 3.11. Its advantage is that it is conceptually and computationally simple and suitable for real-time applications. However, the disadvantage is that it can get stuck in local minima and may require tuning of parameters [22, 23]. An autonomous mobile robot may travel to enter dangerous areas or confined spaces and simultaneously transmit data collected by sensors. It must get to the intended location without running into any other robots through the use of infrared, ultrasonic sensors, or antennas for collision avoidance [24–26].
2. **Discrete Potential Fields:** Divides the environment into a grid and assigns potentials to each cell based on proximity to obstacles and the goal. An example of this is shown in Figure 3.12. Its advantage is that it avoids local minima by discretizing the space, relatively easy to implement. However, the disadvantage is that its accuracy is affected by Grid resolution and may not scale well to complex environments.
3. **Voronoi Diagrams:** Constructs a graph where each vertex represents a point equidistant to two or more obstacles, and edges represent paths. An example of this is shown in Figure 3.13. Its advantage is that it guarantees a path that avoids obstacles, efficient for environments with few obstacles. However, the disadvantage is that its complexity increases with the number of obstacles and may not handle dynamic environments well.
4. **Visibility Graphs:** Connects the vertices of obstacles with straight lines to create a graph, where edges represent paths free of obstacles. An example of this is shown in Figure 3.14. Its advantage is that it finds optimal paths in terms of distance and works well for environments with many obstacles. However, the disadvantage is that it is

computationally intensive for complex environments and requires efficient line-of-sight algorithms.

Figure 3.11 (a) Antigradient of attractive potential. (b) Antigradient of repulsive potential. (c) Antigradient of total potential. (d) Path using artificial potential fields.

Figure 3.12 (a) Discrete potential approximation. (b) Path with discrete potential fields.

Figure 3.13 Path with voronoi diagram.

Controller Block

Trajectory tracking in mobile robotics involves designing control laws to steer a robot along a desired path while avoiding obstacles. The choice of control law depends on the system's dynamics, the desired performance, and the complexity of the environment. In this section, we are concerned with designing appropriate control inputs, v and ω, to the robot to guide its pose toward a given target trajectory or state. "Robot pose" refers to the spatial configuration and orientation of a robot at a given instant. Two types of control activities are discussed in this research: trajectory tracking and posture regulation.

Figure 3.14 Path with visibility graphs.

This block designs a control law to track the trajectory obtained from the planner, with options for choosing linear control, nonlinear control, or input-output linearization.

3.4.4 Trajectory Tracking

The task of trajectory tracking is to follow a specific trajectory over time, ensuring a desired evolution of the robot's pose. In the first instance, we are concerned with reporting the reference trajectory in the form of an objective function. After defining functions $x_{star}(t)$, $y_{star}(t)$, and $\theta_{star}(t)$, which are crucial for further calculations, we present the new parameters for the initial position and angle of the robot. These parameters are

specified because the robot may be positioned slightly differently from the default starting point.

It is important to note that in this phase, the goal is not to precisely reach the target point, but rather to arrive in its vicinity. Additionally, the user can choose one of the three control laws implemented to track the objective function.

Linear Control: It uses a linear feedback law to track the trajectory, where the control inputs are linear functions of the tracking error. It guides a mobile robot along a predefined trajectory. This control law is based on a linearized approximation of the error dynamics around the desired trajectory. The state of the robot $X = [x,y,\theta]$, consisting of its position x,y and orientation θ, is compared to the desired trajectory, defined by functions $x_{star}(t)$, $y_{star}(t)$, and $\theta_{star}(t)$, along with their derivatives. Error terms e_x, e_y, and e_θ are calculated to represent the deviation of the robot from the desired trajectory, considering its current orientation.

(3.21)

(3.22)

(3.23)

(3.24)

(3.25)

where

(3.26)

Figure 3.15 Trajectory tracking with linear control.

Control signals v (linear velocity) and ω (angular velocity) are then computed based on these error terms and the desired trajectory's derivatives. Linear velocity v is the sum of the desired linear velocity along trajectory v_{star} and correction term $k_1 e_x$ that adjusts for positional errors. Angular velocity ω is determined by the desired angular acceleration ω_{star} and

correction terms $k_2 e_y$ and $k_3 e_\theta$ that correct for orientation errors. Parameters a and δ are used for linearization and tuning, respectively, affecting the control law's response. The function returns the time derivative of the robot's state, including the calculated linear and angular velocities, to guide the robot toward following the desired trajectory accurately. Overall, it implements a control law that adjusts the robot's velocities based on its position relative to a desired trajectory, aiming to minimize the tracking error and achieve accurate trajectory tracking.

Figure 3.15 shows trajectory generated by artificial potential fields; it is possible to notice how the path followed by the robot converges with the reference one. In the same way, the values of the variables x, y and θ, respectively position and orientation of the robot, converge with the corresponding reference variables, while the following error tends to vanish as shown in Figure 3.16.

Nonlinear Control: It employs nonlinear functions to calculate the control inputs, allowing for more complex control strategies. In this case, no linearization step is performed.

(3.27)

With $k_2 > 0$, $k_1(v_{star}, \omega_{star}) > 0$, and $k_3(v_{star}, \omega_{star}) > 0$, both bounded with bounded derivatives. Figure 3.17 shows the trajectory generated through artificial potential fields, after a short transient period, the trajectory of the robot converges with the reference one. Also, in this case, all the variables converge to the desired values and the error tends to vanish, as shown in Figure 3.18.

Figure 3.16 Trend of variables x , y , and θ and their error with linear control.

Figure 3.17 Trajectory tracking with nonlinear control.

Input-Output Linearization: This technique, also known as feedback linearization, looks for a nonlinear control law based on virtual inputs such that the relation between these virtual inputs and the system outputs is

linear. It transforms the system dynamics into a linear form by canceling nonlinear terms, simplifying the control design.

In the case of the robot, point B is considered at distance $|b|$ from the center of the robot. It will be point B that follows the reference trajectory, while the robot "follows" B. All this is illustrated in Figure 3.19, where the trajectory is represented by the dotted line, while the black dot is B and the triangle is the robot.

Figure 3.18 Trend of variables x , y , and θ and their error with nonlinear control.

The control law is as follows:

(3.28)

Figure 3.19 Point B that follows the reference trajectory, while the robot follows B.

Figure 3.20 shows trajectory generated by artificial potential fields. In the same way, Figure 3.21 shows the values of variables x, y, and θ, respectively, position and orientation of the robot, converge with the corresponding reference variables, while the following error tends to vanish.

In this case, the position error is never zero. This occurs due to the nature of this law: since the robot does not directly follow the trajectory but instead tracks point B, the error is precisely caused by distance $|b|$ between the robot's center and B.

Figure 3.20 Trajectory tracking with input-output linearization.

Figure 3.21 Trend of variables x , y , and θ and their error with input-output linearization.

3.4.5 Posture Regulation

This block implements a control law for posture regulation, ensuring that

the robot reaches the desired goal orientation. To bring the robot into the "goal" pose defined by the user in the initial phase. The following control law applies:

(3.29)

With $k_1 > 0$, $k_2 > 0$, , and

(3.30)

Figure 3.22 shows the trajectory generated by the controller to go from the arrival point of the trajectory generated in the trajectory tracking phase to the goal point decided by the user. The trend of the variables is shown in Figure 3.23.

Figure 3.22 Posture regulation with artificial potential fields and linear control.

Figure 3.23 Trend of the variables.

3.5 Conclusion

This work presented a valuable opportunity to develop new skills in industrial and mobile robotics. Analyzing the structure of an anthropomorphic robot in industrial robotics and designing a robot capable of autonomous movement in a predefined environment in mobile robotics provided a platform to address technical and conceptual challenges. By developing algorithms and simulating specific trajectories, the project demonstrated the potential of anthropomorphic robots in automating industrial activities.

Similarly, the mobile robot, equipped with navigation algorithms, exhibited the capability to navigate safely and efficiently within the predefined

environment. This project enhanced the understanding and expertise in robotics, laying a foundation for future research and development in the field. The acquired experience serves as a stepping stone for exploring future applications and benefits of anthropomorphic and mobile robots across various sectors.

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Note

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